

Retrieval Augmented Generation

DS205.3 – Data Science in Python

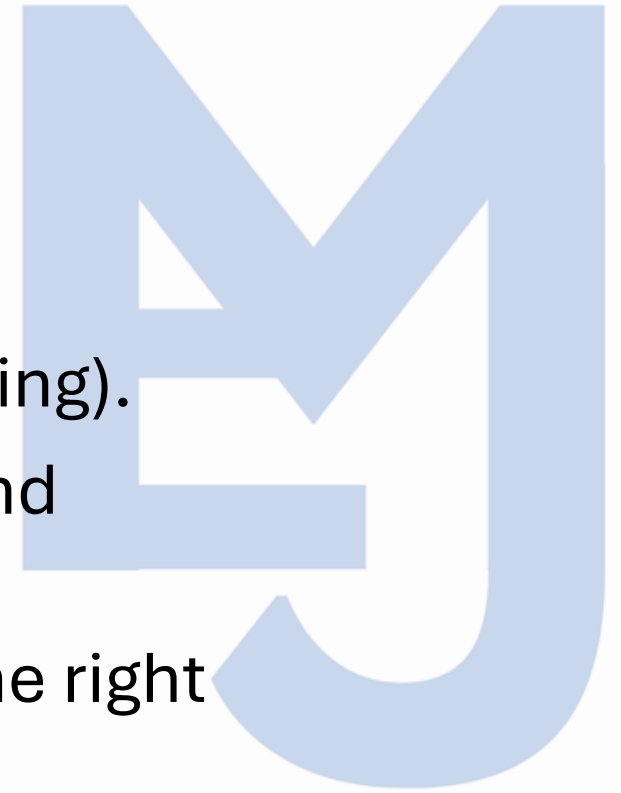
Mr. Anton Jayakody

MSc (CS)[R], BSc (CS)



So far:

- Week 1: Data engineering basics (asyncio, type hinting).
- Week 2: Agentic architecture (OOP, composition, and dependency injection).
- Week 3: Tool routing (The Router pattern to select the right capability).
- Week 4: Prompt Engineering

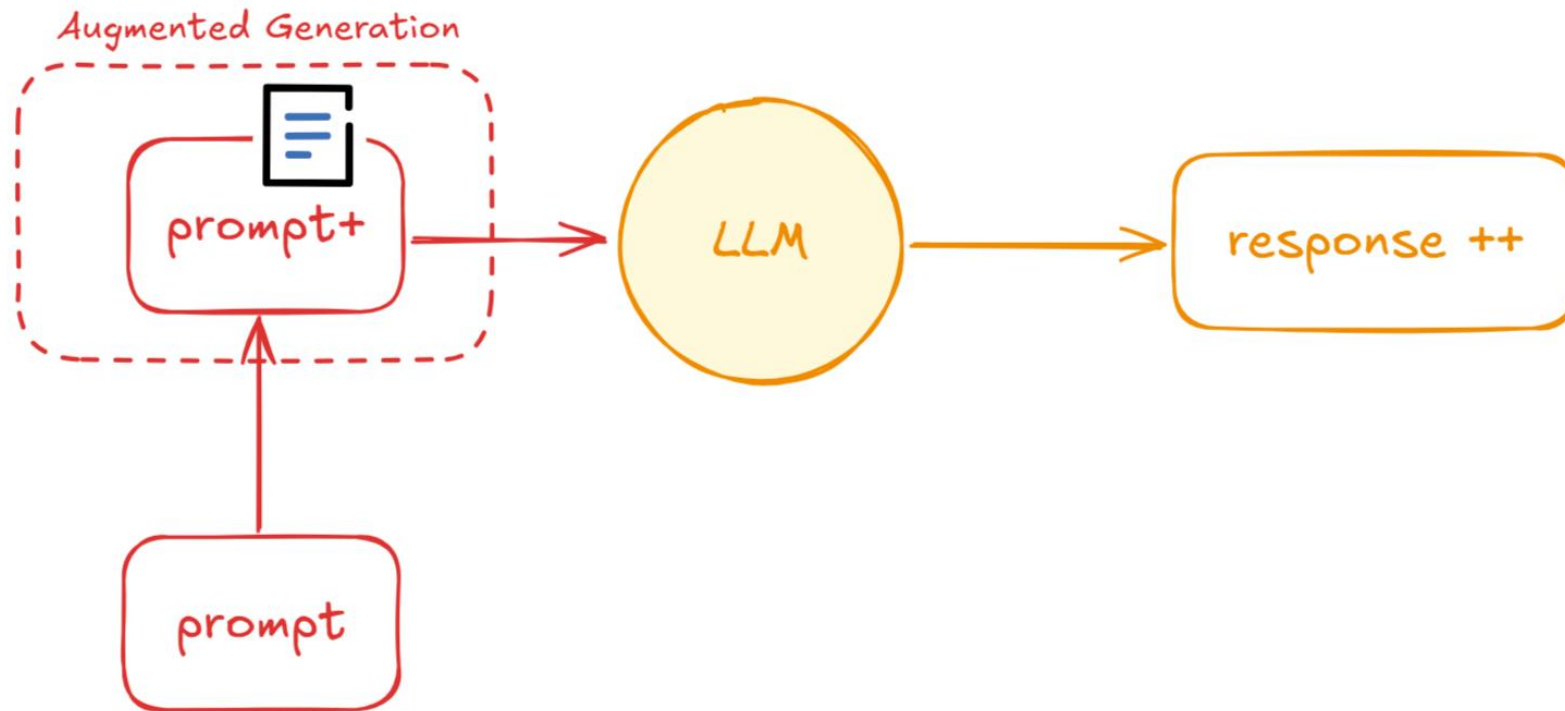


Drawbacks of LLMs

- Hallucination
- Outdated information
- Low efficiency in parameterizing knowledge
- Lack of in-depth knowledge in specialized domains
- Weak inferential capabilities



Augmented Generation



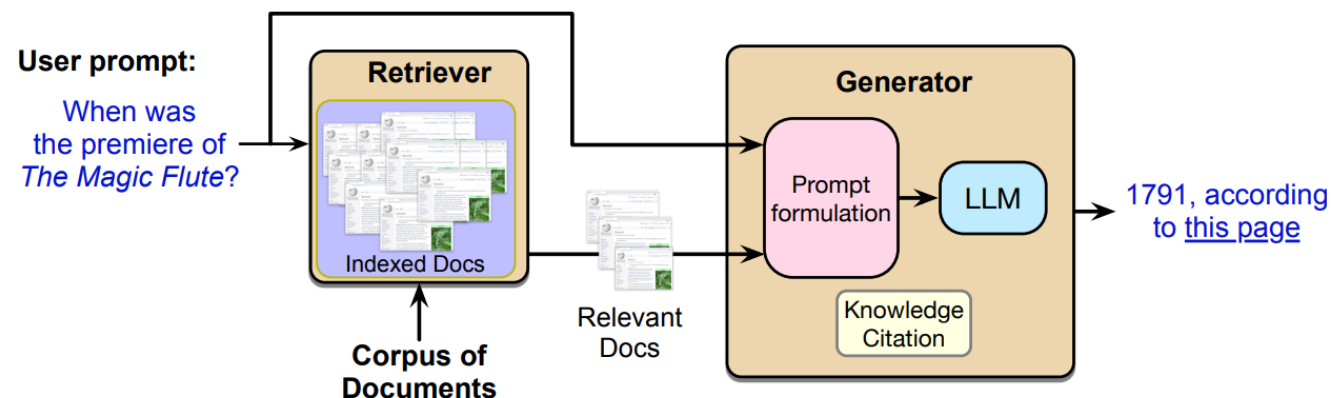
Advantages of Augmented Generation

- Domain-specific accurate answering
- Frequent updates of data
- Traceability and explainability of generated content
- Controllable Cost
- Privacy protection of data



Retrieval-Augmented Generation (RAG)

- When answering questions or generating text, it first retrieves relevant information from a large number of documents, and then LLMs generates answers based on this information.
- By attaching a external knowledge base, there is no need to retrain the entire large model for each specific task.
- The RAG model is especially suitable for knowledge-intensive tasks.



The RAG System

- A RAG system is based on two major components: the retriever and the generator.
- First Retrieve stage, we retrieve relevant passages from some prespecified text collection, for example using the dense retrievers.
- second generate stage, we take the set of retrieved passages, integrate it with the user prompt, and pass some version of these to a large language model to generate an answer

Basic RAG / Naïve RAG

Step1 Indexing

- 1. Divide the document into even chunks, each chunk being a piece of the original text.
- 2. Using the encoding model to generate an embedding for each chunk.
- 3. Store the Embedding of each block in the vector database.

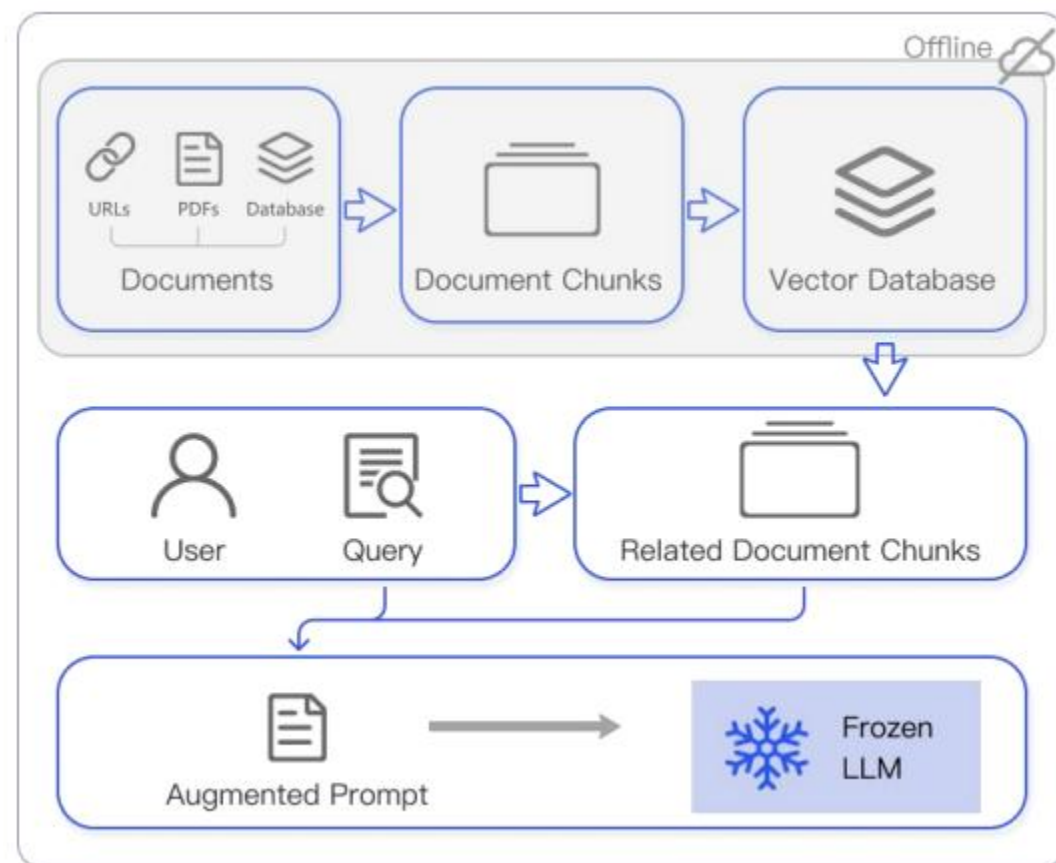
Step2 Retrieval

- Retrieve the k most relevant documents using vector similarity search.

Step3 Generation

- The original query and the retrieved text are combined and input into a LLM to get the final answer

Basic RAG / Naïve RAG Architecture and Applications



Q&A

RETRO (Borgeaud et al, 2021)
REALM (Gu et al, 2020)
ATLAS (Izacard et al, 2023)

Fact Checking

RAG (Lewis et al, 2020)
ATLAS (Izacard et al, 2022)
Evi. Generator (Asai et al, 2022a)

Dialog

BlenderBot3 (Shuster et al, 2022)
Internet-augmented generation (Komeili et al., 2022)

Summary

FLARE (Jiang et al, 2023)

Machine Translation

kNN-MT (Khandelwal et al., 2020)
TRIME-MT (Zhong et al., 2022)

Code Generation

DocPrompting (Zhou et al., 2023)
Natural Prover (Welleck et al., 2022)

Natural Language Inference

kNN-Prompt (Shi et al., 2022)
NPM (Min et al., 2023)

Sentiment analysis

kNN-Prompt (Shi et al., 2022)
NPM (Min et al., 2023)

Commonsense reasoning

Raco (Yu et al, 2022)

Feature Comparison	RAG	Fine-Tuning
Knowledge Updates	Directly updating the retrieval knowledge base ensures that the information remains current without the need for frequent retraining, making it well-suited for dynamic data environments.	Stores static data, requiring retraining for knowledge and data updates.
External Knowledge	Proficient in leveraging external resources, particularly suitable for accessing documents or other structured/unstructured databases.	Can be utilized to align the externally acquired knowledge from pretraining with large language models, but may be less practical for frequently changing data sources.
Data Processing	Involves minimal data processing and handling.	Depends on the creation of high-quality datasets, and limited datasets may not result in significant performance improvements.
Model Customization	Focuses on information retrieval and integrating external knowledge but may not fully customize model behavior or writing style.	Allows adjustments of LLM behavior, writing style, or specific domain knowledge based on specific tones or terms.
Interpretability	Responses can be traced back to specific data sources, providing higher interpretability and traceability.	Similar to a black box, it is not always clear why the model reacts a certain way, resulting in relatively lower interpretability.
Computational Resources	Depends on computational resources to support retrieval strategies and technologies related to databases. Additionally, it requires the maintenance of external data source integration and updates.	The preparation and curation of high-quality training datasets, defining fine-tuning objectives, and providing corresponding computational resources are necessary.
Latency Requirements	Involves data retrieval, which may lead to higher latency.	LLM after fine-tuning can respond without retrieval, resulting in lower latency.
Reducing Hallucinations	Inherently less prone to hallucinations as each answer is grounded in retrieved evidence.	Can help reduce hallucinations by training the model based on specific domain data but may still exhibit hallucinations when faced with unfamiliar input.
Ethical and Privacy Issues	Ethical and privacy concerns arise from the storage and retrieval of text from external databases.	Ethical and privacy concerns may arise due to sensitive content in the training data.

RAG vs Finetuning

RAG in the Big Picture

